

Projectile motion

Part A B C

3unit

selective

Q 1

A projectile has the equation of path $y = -3x^2 + 2x + 4$. How far will it have travelled horizontally before it returns to its original height?

- A. $\frac{1}{3}$ unit B. $\frac{2}{3}$ unit C. 2 units D. 4 units

original height when $y = 4$

$$\begin{aligned} -3x^2 + 2x + 4 &= 4 \\ -3x^2 + 2x &= 0 \\ -x(3x - 2) &= 0 \\ x &= 0 \quad \text{or} \quad x = \frac{2}{3} \end{aligned}$$

Q 2

Zara is testing a batch of homemade bullets on her shooting range.

She holds her rifle at shoulder height (170 cm above the ground) and shoots horizontally with an initial speed of 360 m/s at a target 100m away.

Assume acceleration due to gravity is 10m/s^2 .

- (i) Find expressions for $\dot{y}$ and $\ddot{x}$. 2
 (ii) Find expressions for y and x . 2
 (iii) Hence find the equation of path as y in terms of x . 1
 (iv) Assuming the ground is horizontal at her range, how far above the ground will the bullet hit the target? Give your answer correct to the nearest cm. 1
 (v) When does the bullet hit her target? Find the impact speed (correct to 5 significant figures). 2

(v) from ① above, when $x = 100$,

$$100 = 360t$$

$$\therefore t = \frac{100}{360} = \frac{5}{18} \text{ second} \checkmark$$

$$\text{when } t = \frac{5}{18}, \dot{x} = 360, \dot{y} = -10 \times \frac{5}{18} = -\frac{50}{18}$$

$$\text{speed} = \sqrt{\dot{x}^2 + \dot{y}^2}$$

$$= \sqrt{360^2 + \left(-\frac{50}{18}\right)^2}$$

$$= 360.01 \text{ m/s (5 sig figs)} \checkmark$$

$$(i) \quad \ddot{x} = 0$$

$$\dot{x} = \int 0 dt$$

$$= C_1$$

$$\text{when } t = 0, \dot{x} = 360$$

$$\therefore \dot{x} = 360 \checkmark$$

$$\ddot{y} = -10$$

$$\dot{y} = \int -10 dt$$

$$= -10t + C_2$$

$$\text{when } t = 0, \dot{y} = 0$$

$$\therefore \dot{y} = -10t \checkmark$$

$$ii) \text{ when } t = 0, x = 0$$

$$x = \int 360 dt$$

$$= 360t + C_3$$

$$\text{when } t = 0, x = 0 \therefore C_3 = 0$$

$$\text{and } x = 360t \quad \textcircled{1} \checkmark$$

$$y = \int (-10t) dt$$

$$= -5t^2 + C_4$$

$$\text{when } t = 0, y = 1.7 \text{ m}$$

$$\therefore C_4 = 1.7$$

$$\text{and } y = -5t^2 + 1.7 \quad \textcircled{2} \checkmark$$

$$iii) \text{ from } \textcircled{1} \quad t = \frac{x}{360} \quad \textcircled{3} \quad \text{sub } \textcircled{3} \text{ into } \textcircled{2}:$$

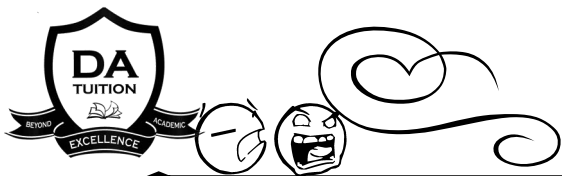
$$y = -5\left(\frac{x}{360}\right)^2 + 1.7$$

$$= -\frac{x^2}{25920} + 1.7 \quad \textcircled{4} \checkmark$$

$$iv) \text{ when } x = 100, \quad y = \frac{-100^2}{25920} + 1.7$$

$$= 1.31 \text{ m (nearest cm)}$$





Projectile motion

Part A B C



Q 3

B
A
U
L
K
H
A
M

A ball is struck by a racquet from a point A which has position vector $21\mathbf{j}$ metres relative to a fixed origin O . Initially, the velocity vector of the ball is $(5\mathbf{i} + 8\mathbf{j})$ m/s, where $\mathbf{i}$ and $\mathbf{j}$ are the unit vectors in the horizontal and vertical directions respectively.

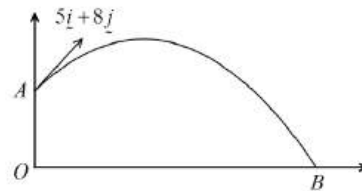
After being struck, the ball undergoes projectile motion under the influence of gravity only, that is, the acceleration vector of the ball $\mathbf{a} = -10\mathbf{j}$. The ball will hit the ground at point B , a point that is on the same horizontal level as the origin.

- (i) Show that the velocity vector of the ball t seconds after being struck is $\mathbf{v} = 5\mathbf{i} + (8 - 10t)\mathbf{j}$. 2

- (ii) Find an expression for the position vector $\mathbf{r}$ of the ball relative to O at a time t seconds after it was struck. 2

- (iii) Find the maximum height that the ball reaches. 2

- (iv) Find the speed of the ball when it lands at B and the angle at which the ball strikes the ground. 3



(i) $\mathbf{a} = -g\mathbf{j}$
 $= -10\mathbf{j}$
 $\mathbf{v} = 5\mathbf{i} - (10t + c_1)\mathbf{j}$
 (since horizontal velocity is a constant 5 m/s)
 Given $\mathbf{v}(0) = 5\mathbf{i} + 8\mathbf{j}$:
 $-(10 \times 0 + c_1) = 8$
 $c_1 = -8$

$\mathbf{v} = 5\mathbf{i} - (10t - 8)\mathbf{j}$
 $= 5\mathbf{i} + (8 - 10t)\mathbf{j}$

(ii) $\mathbf{v} = 5\mathbf{i} + (8 - 10t)\mathbf{j}$
 $\mathbf{r} = (5t + c_2)\mathbf{i} + (8t - 5t^2 + c_3)\mathbf{j}$

Given $\mathbf{r}(0) = 21\mathbf{j}$:

$5 \times 0 + c_2 = 0$ $8 \times 0 - 5 \times 0^2 + c_3 = 21$
 $c_2 = 0$ $c_3 = 21$

$\mathbf{r} = (5t)\mathbf{i} + (8t - 5t^2 + 21)\mathbf{j}$

(iii) Maximum height occurs when $\dot{y}(t) = 0$:
 $8 - 10t = 0$
 $t = 0.8$

$y(0.8) = 8 \times 0.8 - 5 \times 0.8^2 + 21$
 $= 24.2$ metres

(iv) When the ball will lands, $y(t) = 0$:
 $-5t^2 + 8t + 21 = 0$
 $(3 - t)(5t + 7) = 0$
 $t = 3$ (since $t \geq 0$)

$\mathbf{v}(3) = 5\mathbf{i} + (8 - 10 \times 3)\mathbf{j}$
 $= 5\mathbf{i} - 22\mathbf{j}$

Speed at $B = \sqrt{5^2 + 22^2}$
 $= 22.56$ m/s (2 d.p.)

Angle of impact $= \tan^{-1}\left(\frac{22}{5}\right)$
 $= 77^\circ$ (nearest degree)

(Also accept the obtuse angle 103° and disregard whether they state "above" or "below the horizontal")

Q 4

B
O
N
N
Y
R
I
G
G

A particle is fired from the origin O with initial velocity 20 ms^{-1} at an angle 30° to the horizontal.

The equations of motion are: $\frac{d^2x}{dt^2} = 0$ and $\frac{d^2y}{dt^2} = -10$

- (i) Show that $x = 10\sqrt{3}t$. 1

- (ii) Show that $y = 10t - 5t^2$. 2

- (iii) Hence find the Cartesian equation for the trajectory of the particle. 1

(i) $\frac{d^2x}{dt^2} = 0$

$\frac{dx}{dt} = 20 \times \cos 30^\circ = 20 \times \frac{\sqrt{3}}{2} = 10\sqrt{3}$

$x = 10\sqrt{3}t + C_1$

$x(0) = 0 \rightarrow C_1 = 0$

$\therefore x = 10\sqrt{3}t$

(ii) $\frac{d^2y}{dt^2} = -10$

$\frac{dy}{dt} = -10t + C_2$

$\frac{dy}{dt}|_{t=0} = 20 \times \sin 30^\circ = 10 \rightarrow C_2 = 10$

$\therefore \frac{dy}{dt} = -10t + 10$

$y = -5t^2 + 10t + C_3$

$y(0) = 0 \rightarrow C_3 = 0$

$\therefore y = -5t^2 + 10t$

(iii) Re-arranging $x = 10\sqrt{3}t$, we obtain: $t = \frac{x}{10\sqrt{3}}$

Substituting this into the equation in y :

$y = 10\left(\frac{x}{10\sqrt{3}}\right) - 5\left(\frac{x}{10\sqrt{3}}\right)^2$

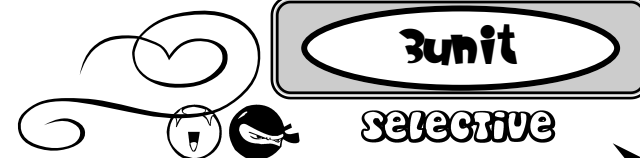
$\therefore y = \frac{x}{\sqrt{3}} - \frac{x^2}{60}$





Projectile motion

Part A B C



Q 5

C
A
R
I
N
G
B
A
H

A vertical tower of height h metres stands with its base at a point O on horizontal ground. At the same instant, stone A is projected from O with speed $V \text{ ms}^{-1}$ at an angle α above the horizontal and stone B is projected from the top of the tower with speed $U \text{ ms}^{-1}$ at an angle β above the horizontal, where $\alpha > \beta$. The two stones move in the same vertical plane under gravity, where the acceleration due to gravity is $g \text{ ms}^{-2}$, and collide after T seconds. At time t seconds after firing, the position vectors of the two stones are

$$r_A(t) = (Vt \cos \alpha)\mathbf{i} + (-\frac{1}{2}gt^2 + Vt \sin \alpha)\mathbf{j}$$

$$r_B(t) = (Ut \cos \beta)\mathbf{i} + (h - \frac{1}{2}gt^2 + Ut \sin \beta)\mathbf{j} \quad (\text{DO NOT PROVE THIS})$$

Show that $T = \frac{h \cos \alpha}{U \sin(\alpha - \beta)}$.

(b) At time T , $r_A(T) = r_B(T)$

Hence $VT \cos \alpha = UT \cos \beta$ (1)

$-\frac{1}{2}gT^2 + VT \sin \alpha = h - \frac{1}{2}gT^2 + UT \sin \beta$ (2)

∴ (1) $\Rightarrow V \cos \alpha = U \cos \beta$ (3)

(2) $\Rightarrow VT \sin \alpha = h + UT \sin \beta$ (4)

(3) $\times T \sin \alpha$

$VT \sin \alpha \cos \alpha = UT \cos \beta \sin \alpha$ (5)

(5) $\times \cos \alpha$

$VT \sin \alpha \cos \alpha = h \cos \alpha + UT \sin \beta \cos \alpha$ (6)

(6) - (5)

$h \cos \alpha + UT \sin \beta \cos \alpha - UT \cos \beta \sin \alpha = 0$

$UT (\sin \beta \cos \alpha - \cos \beta \sin \alpha) = -h \cos \alpha$

$UT (\sin \alpha \cos \beta - \sin \beta \cos \alpha) = h \cos \alpha$

$UT \sin(\alpha - \beta) = h \cos \alpha$

∴ $T = \frac{h \cos \alpha}{U \sin(\alpha - \beta)}$

Q 6

C
H
E
L
T
E
N
H
A
M

G
I
R
L
S

A golf ball is hit at a velocity of 110 ms^{-1} at an angle θ to the horizontal.

The position vector $s(t)$, from the starting point, of the ball after t seconds is given by

$$s = 110t \cos \theta \mathbf{i} + (110t \sin \theta - 4.9t^2)\mathbf{j}$$

i) Using gravity of 9.8 ms^{-2} show that the maximum horizontal range of the ball is $\frac{12100 \sin 2\theta}{9.8}$ metres. 2

ii) To ensure that the ball lands on the green, it must travel between 400 and 450 metres. What values of θ , correct to the nearest minute, would allow this to happen? 1

iii) The golfer aims accurately and hits the ball directly towards the green. After 3.4 seconds of flight, at a point 8 metres above the ground, the ball hits a low flying TV drone. If it had not hit the drone or any other obstacles, would the ball have landed on the green? 3

i) Max range when $y = 0$

$$110t \sin \theta - 4.9t^2 = 0$$

$$t(110 \sin \theta - 4.9t) = 0$$

$$t = 0$$

$$\text{or } 110 \sin \theta = 4.9t$$

$$t = \frac{110 \sin \theta}{4.9}$$

When $t = \frac{110 \sin \theta}{4.9}$

$$x = 110 \left(\frac{110 \sin \theta}{4.9} \right) \cos \theta$$

$$= \frac{110^2 \sin \theta \cos \theta}{4.9}$$

$$= \frac{12100 \times 2 \sin \theta \cos \theta}{9.8}$$

$$= \frac{12100 \sin 2\theta}{9.8}$$

ii) We want max range to be between 400 and 450 metres.

$$400 < \frac{12100 \sin 2\theta}{9.8} < 450$$

$$\frac{400 \times 9.8}{12100} < \sin 2\theta < \frac{450 \times 9.8}{12100}$$

$$18^\circ 54' 10.8'' < 2\theta < 21^\circ 22' 28.3''$$

$$9^\circ 27' < \theta < 10^\circ 41'$$

iii) When $t = 3.4$ sec, $y = 8$ m

To find the angle the ball was hit at:

$$y = 110t \sin \theta - 4.9t^2$$

$$8 = 110(3.4) \sin \theta - 4.9(3.4)^2$$

$$\sin \theta = \frac{8 + 4.9(3.4)^2}{110(3.4)}$$

$$\theta = 9^\circ 57' 11.77''$$

To find the distance travelled using this angle, using time of flight from (a):

$$t = \frac{110 \sin \theta}{4.9}$$

$$= \frac{110 \sin 9^\circ 57''}{4.9}$$

$$= 3.880192077 \text{ sec}$$

$$x = 110 \times 3.880192077 \cos 9^\circ 57' 11.77''$$

$$= 420.3970664 \text{ m}$$

Therefore, if the ball hadn't hit the drone, it would have made the green.





Projectile motion

Part A B C



Q 7

Jerry stands at the top of a building 30 m tall and throws a ball with a velocity of 15 m s^{-1} at an angle of 45° to the horizontal. The ball eventually reaches the ground.

C
S
S
A

- (i) Derive the vector displacement of the ball in the form $\underline{s} = x\underline{i} + y\underline{j}$. You may assume that $g = 10 \text{ m s}^{-2}$.
- (ii) Find the time taken for the ball to reach the ground.

Find the components of displacement

$$\ddot{y} = -g$$

$$\dot{y} = -gt + c_1$$

$$\text{when } t = 0, \dot{y} = \frac{15\sqrt{2}}{2} \therefore c_1 = \frac{15\sqrt{2}}{2}$$

$$\therefore \dot{y} = -gt + \frac{15\sqrt{2}}{2}$$

$$y = -\frac{gt^2}{2} + \frac{15\sqrt{2}}{2}t + c_2$$

$$\text{when } t = 0, y = 30 \therefore c_2 = 30$$

$$\therefore y = -\frac{gt^2}{2} + \frac{15\sqrt{2}}{2}t + 30$$

$$\therefore y = -5t^2 + \frac{15\sqrt{2}}{2}t + 30 \quad \text{using } g = 10$$

$$\therefore \underline{s} = \left(\frac{15\sqrt{2}}{2}t \right) \underline{i} + \left(-5t^2 + \frac{15\sqrt{2}}{2}t + 30 \right) \underline{j}$$

$$\ddot{x} = 0$$

$$\dot{x} = c_3$$

$$\text{when } t = 0, \dot{x} = \frac{15\sqrt{2}}{2} \therefore c_3 = \frac{15\sqrt{2}}{2}$$

$$\therefore \dot{x} = \frac{15\sqrt{2}}{2}$$

$$x = \frac{15\sqrt{2}}{2}t + c_4$$

$$\text{when } t = 0, x = 0 \therefore c_4 = 0$$

$$\therefore x = \frac{15\sqrt{2}}{2}t$$

For the ball to reach the ground the component of $\underline{j}$ must equal zero.

$$-5t^2 + \frac{15\sqrt{2}}{2}t + 30 = 0$$

$$t = \frac{-\frac{15\sqrt{2}}{2} \pm \sqrt{\left(-\frac{15\sqrt{2}}{2}\right)^2 - 4(-5)(30)}}{2(-5)}$$

$$= \frac{-\frac{15\sqrt{2}}{2} \pm \sqrt{712.5}}{-10}$$

$$= 3.7 \text{ seconds (1 dp)} \quad \text{taking } t > 0$$

Q 8

From a cliff 100 m high, a projectile is launched at 90 m/s with an angle of 60° to the horizontal. Let $g = 9.8 \text{ m/s}^2$.

E
N
T
E
R
P
R
I
S
E

- (i) Find the vector for the projectile's displacement in the form

$$\underline{s} = a\underline{i} + b\underline{j}$$

- (ii) Find the time the projectile hits the ground.

(i)

$$a = 9.8 \underline{j}$$

$$\underline{v} = 9.8t \underline{j} + C$$

$$t = 0$$

$$\underline{v} = V \cos \theta \underline{i} + V \sin \theta \underline{j}$$

$$= 90 \cos 60^\circ \underline{i} + 90 \sin 60^\circ \underline{j}$$

$$= 45 \underline{i} + 45\sqrt{3} \underline{j}$$

$$9.8(0) \underline{j} + C = 45 \underline{i} + 45\sqrt{3} \underline{j}$$

$$C = 45 \underline{i} + 45\sqrt{3} \underline{j}$$

$$\underline{v} = 9.8t \underline{j} + 45 \underline{i} + 45\sqrt{3} \underline{j}$$

$$\underline{v} = 45 \underline{i} + (9.8t + 45\sqrt{3}) \underline{j}$$

$$\underline{s} = 45 \underline{i} + (9.8t + 45\sqrt{3}) \underline{j} \, dt$$

$$= 45t \underline{i} + (4.9t^2 + 45\sqrt{3}t) \underline{j} + C_1$$

$$\text{when } t = 0, \underline{s} = 0 \underline{i} + 100 \underline{j} = 100 \underline{j}$$

$$100 \underline{j} = 45(0) \underline{i} + [4.9(0)^2 + 45\sqrt{3}(0)] \underline{j} + C_1$$

$$100 \underline{j} = C_1$$

$$\underline{s} = 45t \underline{i} + (4.9t^2 + 45\sqrt{3}t) \underline{j} + 100 \underline{j}$$

$$\underline{s} = 45t \underline{i} + (4.9t^2 + 45\sqrt{3}t + 100) \underline{j}$$

(ii)

$$4.9t^2 + 45\sqrt{3}t + 100 = 0$$

$$4.9t^2 - 45\sqrt{3}t - 100 = 0$$

$$t = \frac{45\sqrt{3} \pm \sqrt{6075 - 4 \cdot (4.9) \cdot (-100)}}{2 \cdot 4.9}$$

$$= 17.100016...$$

$$= 17.1 \text{ sec}$$



Projectile motion

Part A B C

3unit

selective

Q 9

A cricket ball is hit from an origin at ground level so that its position vector at time t is given by $\mathbf{r}(t) = 15t\mathbf{i} + (20t - 5t^2)\mathbf{j}$ for $t \geq 0$, where $\mathbf{i}$ is the unit vector in the forward direction and $\mathbf{j}$ is a unit vector vertically up.

When the cricket ball reaches its maximum height, its position vector is

- A. $\mathbf{r} = 30\mathbf{i} + 20\mathbf{j}$ C. $\mathbf{r} = 60\mathbf{i}$
B. $\mathbf{r} = 15\mathbf{i} + 20\mathbf{j}$ D. $\mathbf{r} = 30\mathbf{i} + 10\mathbf{j}$

$$\mathbf{r}(t) = 15t\mathbf{i} + (20t - 5t^2)\mathbf{j}$$

$$\text{Max range when } (20t - 5t^2) = 0$$

$$5t(4 - t) = 0$$

$\therefore$ max range occurs when $t = 4$
 $\Rightarrow$ max height when $t = 2$

$\therefore$ Position vector at max height:

$$\mathbf{r}(2) = 30\mathbf{i} + 20\mathbf{j}$$

Q10

A ball is thrown from the origin O with a velocity V and an angle of elevation of θ , where $\theta \neq \frac{\pi}{2}$. What is the Cartesian equation of the flight path? Take $g = 10 \text{ ms}^{-2}$.

- (A) $y = x \tan \theta - \frac{5x^2}{V^2} (1 + \tan^2 \theta)$
(B) $y = x \tan \theta - \frac{10x^2}{V^2} (1 + \tan^2 \theta)$
(C) $x = V \cos \theta t$ and $y = -5t^2 + V \sin \theta t$
(D) $x = V \cos \theta t$ and $y = -10t^2 + V \sin \theta t$

Q11

A golf ball is hit at a velocity of 55 ms^{-1} at an angle θ to the horizontal, with an acceleration due to gravity of 9.8 ms^{-2} being applied to the ball.

The position vector $\mathbf{s}(t)$, from the starting point, of the ball after t seconds is given by

$$\mathbf{s} = 55t \cos \theta \mathbf{i} + (55t \sin \theta - 4.9t^2)\mathbf{j}$$

- i) Show that the ball travels $\frac{3025 \sin 2\theta}{9.8}$ metres before hitting the ground. 2
ii) To ensure that the ball lands on the green, it must travel between 200 and 250 metres. 2
What values of θ , correct to the nearest minute, would allow this to happen?
iii) The golfer aims accurately and hits the ball directly towards the green. 1

After 3.4 seconds of flight, at a point 8 metres above the ground, the ball hits a low flying TV drone.

If it had not hit the drone or any other obstacles, would the ball have landed on the green?

- i) $\mathbf{s}(t) = 55t \cos \theta \mathbf{i} + (55t \sin \theta - 4.9t^2)\mathbf{j}$
when the ball hits the ground:
 $55t \sin \theta - 4.9t^2 = 0$
 $t(55 \sin \theta - 4.9t) = 0$
 $\Rightarrow t = 0, t = \frac{55 \sin \theta}{4.9} \text{ sec}$
 $t = 0$ is at the start of motion
 $\therefore$ distance travelled = $55 \left(\frac{55 \sin \theta}{4.9} \right) \cos \theta$
 $= \frac{3025 \times 2 \sin \theta \cos \theta}{9.8}$
 $= \frac{3025 \sin 2\theta}{9.8} \text{ m}$
ii) $\frac{3025 \sin 2\theta}{9.8} = 200$; $\frac{3025 \sin 2\theta}{9.8} = 250$
 $\Rightarrow \sin 2\theta = \frac{200 \times 9.8}{3025}$; $\sin 2\theta = \frac{250 \times 9.8}{3025}$
 $\Rightarrow \theta \neq 20^\circ 12'$; $\theta = 27^\circ 3'$

Therefore the angle of projection must be $20^\circ 12' \leq \theta \leq 27^\circ 3'$

- iii) substituting $t = 3.4 \text{ sec}$ and 8 for vertical displacement into $55t \sin \theta - 4.9t^2$
 $\Rightarrow 55(3.4) \sin \theta - 4.9(3.4)^2 = 8$
 $\Rightarrow \sin \theta = \frac{8 + 4.9(3.4)^2}{55(3.4)}$
 $\therefore \theta \approx 20^\circ 13'$
Since θ satisfies the condition in part (ii), the ball would have landed on the green.

Projectile motion

Part A B C

Unit

Selective

Q12

A particle is projected across horizontal ground from the origin O . Its initial velocity vector is $12\mathbf{i} + 5\mathbf{j}$ and its acceleration vector is $0\mathbf{i} - 10\mathbf{j}$. Find:

G
O
S
F
O
R
D

- the initial speed of the particle 1
- the angle of projection, correct to the nearest minute 1
- Beginning with its acceleration vector, derive the velocity at any time t in the form $\mathbf{v} = x\mathbf{i} + y\mathbf{j}$ 2
- Show that its displacement at time t is $\mathbf{r} = (12t)\mathbf{i} + (5t - 5t^2)\mathbf{j}$ 1
- Find the maximum height of the particle. 1

d) $\mathbf{v} = 12\mathbf{i} + 5\mathbf{j}$ $\mathbf{a} = 0\mathbf{i} - 10\mathbf{j}$

at $t=0$
 $\mathbf{v} = 12\mathbf{i} + 5\mathbf{j}$ $\mathbf{a} = 0\mathbf{i} - 10\mathbf{j}$

i) at $t=0$
 $|\mathbf{v}| = \sqrt{12^2 + 5^2} = 13 \text{ m/s}$ ①

ii) $\tan \theta = \frac{5}{12}$
 $\therefore \theta = 22^\circ 37'$ ①

iii) $\mathbf{a} = 0\mathbf{i} - 10\mathbf{j}$

$\ddot{x} = 0$ $\ddot{y} = -10$
 $\dot{x} = 0t + c$ $\dot{y} = -10t + c_1$
 at $t=0$ $\dot{x} = 12$, $\dot{y} = 5$

$\therefore c = 12$, $c_1 = 5$
 $\dot{x} = 12$ $\dot{y} = -10t + 5$

$\therefore \mathbf{v} = 12\mathbf{i} + (-10t + 5)\mathbf{j}$ ②

iv)

$x = 12t + c_2$ $y = \frac{-10t^2}{2} + 5t + c_3$

at $t=0$ $x=0$ $y=0$
 $\therefore c_2 = 0 = c_3$

$x = 12t$ $y = -5t^2 + 5t$

$\therefore \mathbf{r} = 12t\mathbf{i} + (5t - 5t^2)\mathbf{j}$ ①

v) max height at $\dot{y} = 0$
 $\therefore -10t + 5 = 0$
 $10t = 5$
 $t = 0.5$

$\therefore y = -5(0.5)^2 + 5(0.5)$
 $= 1.25 \text{ m}$ ①

Q13

A golfer hits a golf ball from a point O with speed V m/s at an angle θ above the horizontal, where $0^\circ < \theta < 90^\circ$. The ball just passes over a 2.25 m high tree after 1.5 seconds. The tree is 60 metres away on level ground from the point from which the ball was hit. The acceleration due to gravity is 10 m/s^2 .

The position vector, $\mathbf{r}$, at any time, t seconds, is given by

$$\mathbf{r}(t) = Vt \cos \theta \mathbf{i} + (Vt \sin \theta - 5t^2) \mathbf{j} \quad (\text{Do NOT prove})$$

- What is the initial angle, θ , of projection of the golf ball, to the nearest minute? 3
- What is the initial speed (V m/s) of the golf ball. 1
- How far from the base of the tree does the ball strike the ground? 1
Leave your answer correct to the nearest metre.

(c) (i)

After 1.5 seconds $x = 60$ and $y = 2.25$

$$60 = 1.5V \cos \theta$$

$$V \cos \theta = 40 \quad (1)$$

$$2.25 = -5 \times 1.5^2 + 1.5V \sin \theta$$

$$13.5 = 1.5V \sin \theta$$

$$V \sin \theta = 9 \quad (2)$$

Dividing the two equations

$$\frac{V \sin \theta}{V \cos \theta} = \frac{9}{40}$$

$$\tan \theta = \frac{9}{40}$$

$$\theta = \tan^{-1} \frac{9}{40} = 12^\circ 41'$$

$\therefore$ Golf ball has an angle of projection of $12^\circ 41'$.

(iii)

$$\mathbf{r}(t) = t \times \underbrace{V \cos \theta}_{40} \mathbf{i} + \left(t \times \underbrace{V \sin \theta}_9 - 5t^2 \right) \mathbf{j}$$

$$= 40t \mathbf{i} + (9t - 5t^2) \mathbf{j}$$

(ii)

Using equations (1) and (2)

$$(V \sin \theta)^2 + (V \cos \theta)^2 = 9^2 + 40^2$$

$$V^2 (\sin^2 \theta + \cos^2 \theta) = 9^2 + 40^2$$

$$V = \sqrt{9^2 + 40^2}$$

$$V = 41 \text{ ms}^{-1}$$

$\therefore$ Speed of the gold ball is 41 ms^{-1}

The ball will land when $9t - 5t^2 = 0 \Rightarrow t = \frac{9}{5} = 1.8$

So the ball lands $40 \times 1.8 = 72 \text{ m}$ from the start which is 12 m from the tree.

Projectile motion

Part A B C

3unit

selective

Q14

KILLAR A

A particle is projected from O , with speed $V \text{ ms}^{-1}$ at an angle of elevation of α to the horizontal on a flat plane.

You may assume the following equations of motion of the projectile:

$$a = -g\mathbf{j}$$

$$\mathbf{v} = (V\cos\alpha)\mathbf{i} + (-gt + V\sin\alpha)\mathbf{j}$$

$$\mathbf{r} = (V\cos\alpha)\mathbf{i} + \left(-\frac{gt^2}{2} + V\sin\alpha\right)\mathbf{j}$$

- (i) Prove that the horizontal range of the projectile is given by

$$\frac{V^2 \sin 2\alpha}{g} \text{ metres}$$

2

A garden sprinkler sprays water symmetrically about a vertical axis at a constant speed $V \text{ ms}^{-1}$. The initial direction of spray varies continuously between angles of 15° and 60° to the horizontal.

- (ii) Prove that, from the fixed position O on the level ground, the sprinkler will wet the surface of an annular region with internal and external radii $\frac{V^2}{2g}$ metres and $\frac{V^2}{g}$ metres respectively.

2

- (iii) Deduce that by locating the sprinkler at O , a rectangular garden bed of size 6m by 3m can be completely watered only if

3

$$\frac{V^2}{2g} \geq 1 + \sqrt{7}$$

(i)

$$x = V \cos \alpha t \rightarrow t = \frac{x}{V \cos \alpha}$$

$$y = -\frac{gt^2}{2} + V \sin \alpha t$$

$$y = -\frac{g}{2} \left(\frac{x}{V \cos \alpha} \right)^2 + V \sin \alpha \frac{x}{V \cos \alpha}$$

Hence, the Cartesian equation is

$$y = x \tan \alpha - \frac{gx^2}{2V^2 \cos^2 \alpha}$$

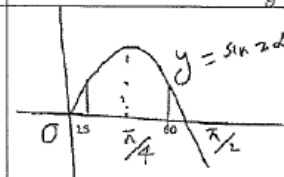
But when $y = 0, x = R$

$$R \tan \alpha = \frac{gR^2}{2V^2 \cos^2 \alpha}$$

$$2V^2 R \sin \alpha \cos \alpha = gR^2$$

$$R \neq 0, R = \frac{V^2 \sin 2\alpha}{g}$$

(ii)



The maximum range is attained when $\alpha = 45^\circ$
The sprinkler water annual region for the angles between 15° and 45°

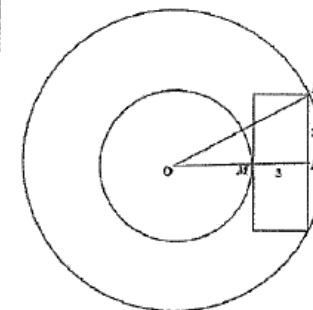
When $\alpha = 15^\circ$,

$$\text{the range } R = \frac{V^2 \sin 30}{g} = \frac{V^2}{2g}$$

When $\alpha = 45^\circ$,

$$\text{the range } R = \frac{V^2 \sin 90}{g} = \frac{V^2}{g}$$

(iii)



In the right-angled triangle ONP,

$$ON^2 = OP^2 - NP^2$$

$$= \left(\frac{V^2}{g} \right)^2 - 9 \quad 1 \text{ mark}$$

The entire garden bed will be wet when

$$ON \geq \frac{V^2}{2g} + 3$$

Thus

$$\left(\frac{V^2}{g} \right)^2 - 9 \geq \left(\frac{V^2}{2g} + 3 \right)^2 \quad 1 \text{ Mark}$$

$$\frac{V^4}{g^2} - 9 \geq \frac{V^4}{4g^2} + \frac{3V^2}{g} + 9$$

$$\frac{3V^4}{4g^2} - \frac{3V^2}{g} - 18 \geq 0$$

$$\frac{V^4}{4g^2} - \frac{V^2}{g} \geq 6$$

$$\left(\frac{V^2}{2g} - 1 \right)^2 \geq 7$$

$$\frac{V^2}{2g} \geq 1 \pm \sqrt{7}$$

$$\frac{V^2}{2g} > 0$$

Hence,

$$\frac{V^2}{2g} > 1 + \sqrt{7}$$

1 mark



Projectile motion

Part A B C



Q15

NEAP

A particle is projected from a point O on level horizontal ground with a speed of 21 m s^{-1} at an angle θ to the horizontal. At time T seconds, the particle passes through the point $B(12, 2)$.

Neglecting the effects of air resistance, the equations describing the motion of the particle are:

$$x = Vt \cos \theta$$

$$y = Vt \sin \theta - \frac{1}{2}gt^2$$

where t is the time in seconds after projection, $g \text{ m s}^{-2}$ is the acceleration due to gravity where $g = 9.8 \text{ m s}^{-2}$ and x and y are measured in metres. Do NOT prove these equations.

- | | |
|---|---|
| (i) By considering the horizontal component of the particle's motion, show that $T = \frac{4}{7} \sec \theta$. | 1 |
| (ii) By considering the vertical component of the particle's motion and, using the result from part (a) (i), show that $4 \tan^2 \theta - 30 \tan \theta + 9 = 0$. | 2 |
| (iii) Find the particle's least possible flight time from O to B . Give your answer correct to two decimal places. | 2 |



- (i) Substituting $x = 12$, $V = 21$ and $t = T$ into

$$x = Vt \cos \theta \text{ gives } 12 = 21T \cos \theta \Rightarrow T = \frac{12}{21 \cos \theta}.$$

$$\text{Cancelling and using } \frac{1}{\cos \theta} = \sec \theta \text{ gives } T = \frac{4}{7} \sec \theta.$$

- (ii) Substituting $y = 2$, $V = 21$ and $t = T$ into

$$y = Vt \sin \theta - \frac{1}{2}gt^2 \text{ gives } 2 = 21T \sin \theta - 4.9T^2.$$

$$\text{Substituting } T = \frac{4}{7} \sec \theta \text{ into}$$

$$2 = 21T \sin \theta - 4.9T^2 \text{ gives:}$$

$$2 = 21 \left(\frac{4}{7} \sec \theta \right) \sin \theta - 4.9 \left(\frac{4}{7} \sec \theta \right)^2$$

$$= 12 \tan \theta - \frac{8}{5} \sec^2 \theta$$

$$= 12 \tan \theta - \frac{8}{5} (1 + \tan^2 \theta)$$

$$10 = 60 \tan \theta - 8(1 + \tan^2 \theta)$$

$$0 = 8 \tan^2 \theta - 60 \tan \theta + 18$$

$$\text{So } 4 \tan^2 \theta - 30 \tan \theta + 9 = 0.$$

- (iii) Using the quadratic formula to solve

$$4 \tan^2 \theta - 30 \tan \theta + 9 = 0 \text{ for } \tan \theta \text{ gives}$$

$$\tan \theta = \frac{15 \pm 3\sqrt{21}}{4} (= 0.3130\dots, 7.1869\dots).$$

The shortest flight time occurs for

$$\theta = \tan^{-1} \left(\frac{15 - 3\sqrt{21}}{4} \right) (= 0.3130\dots).$$

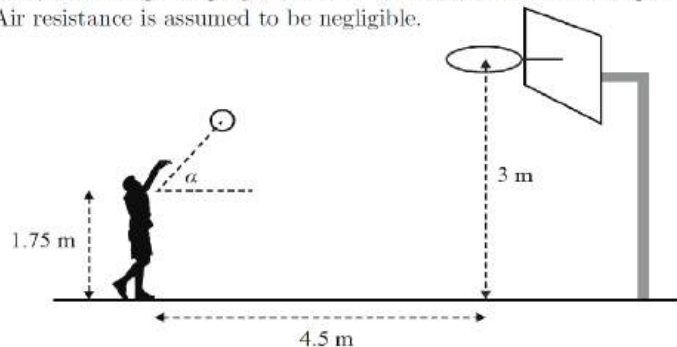
$$\text{Substituting } \theta = \tan^{-1} \left(\frac{15 - 3\sqrt{21}}{4} \right) (= 0.3130\dots) \text{ into}$$

$$T = \frac{4}{7} \sec \theta \text{ gives } T = 0.60 \text{ (s).}$$

Q16

NORMANHURST

A basketball player aims to throw a basketball through a ring, the centre of which is at a horizontal distance of 4.5 m from the point of release of the ball and 3 m above floor level. The ball is released at a height of 1.75 m above floor level, at an angle of projection α to the horizontal and at a speed of $V \text{ ms}^{-1}$. Air resistance is assumed to be negligible.



The position vector of the centre of the ball at any time t seconds, for $t \geq 0$, relative to the point of release is given by

$$\mathbf{r} = \begin{pmatrix} Vt \cos \alpha \\ Vt \sin \alpha - 5t^2 \end{pmatrix}$$

Displacement components are measured in metres. For the player's first shot at goal, $V = 7 \text{ ms}^{-1}$ and $\alpha = 45^\circ$.

- Find the time, in seconds, taken for the ball to reach its maximum height. 2
Give your answer in the form $\frac{a\sqrt{b}}{c}$, where a, b and c are positive integers.
- Find the maximum height, in metres, above floor level, reached by the centre of the ball. 2
Give your answer correct to two decimal places.
- Find the distance between the centre of the ball and the centre of the ring when the ball reaches its maximum height. 2
Give your answer in metres, correct to two decimal places.

$$\text{i. } y = \left(d \frac{7}{\sqrt{2}} - 10t \right)$$

$$\text{When } \frac{7}{\sqrt{2}} - 10t = 0,$$

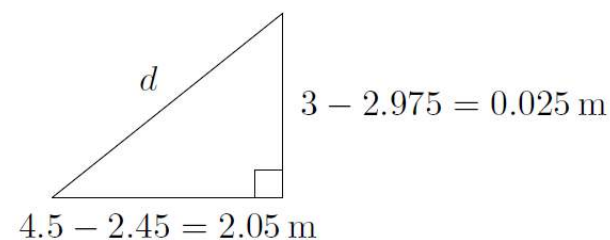
$$t = \frac{7}{10\sqrt{2}} = \frac{7\sqrt{2}}{20}$$

$$\text{ii. } y = \frac{7}{\sqrt{2}} \times \frac{7\sqrt{2}}{20} - 5 \left(\frac{7\sqrt{2}}{20} \right)^2 = 1.225$$

$$\therefore 1.225 + 1.75 \approx 2.98 \text{ m}$$

$$\text{iii. } x = \frac{7}{\sqrt{2}} \times \frac{7\sqrt{2}}{20} = \frac{49}{20} = 2.45 \text{ m}$$

Let d be the distance between the centre of the ball and the centre of the ring.



$$d = \sqrt{2.05^2 + 0.025^2} = 2.05 \text{ m}$$

Projectile motion

Part A B C

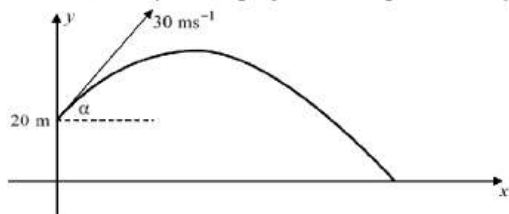
3unit

selective

Q17

A projectile is fired from a point 20 metres above level ground, with a velocity of 30 ms^{-1} and at an angle of α° above the horizontal.

The displacement, velocity and acceleration of the projectile at any time are given by the displacement vector $\underline{r}$, the velocity vector $\underline{v}$ and the acceleration vector $\underline{a}$ respectively. The initial velocity of the projectile is represented by the vector $\underline{u}$.



The acceleration vector is $\underline{a} = -10\hat{j}$ and has units of ms^{-2} .

- Write down, in component form, the initial velocity vector $\underline{u}$.
- Derive the displacement vector $\underline{r}$ in component form.
- By showing that the Cartesian equation of the trajectory is

$y = x \tan \alpha - \frac{x^2}{180} \sec^2 \alpha + 20$, or otherwise, find the value(s) of α for which the projectile will hit a target 90 metres down-range and 1 metre above the ground.

(i) $\underline{u} = 30 \cos \alpha \hat{i} + 30 \sin \alpha \hat{j}$ [1]

(ii) $\underline{a} = -10\hat{j}$
 $\underline{v} = -10t\hat{j} + \underline{c}_1$

At $t=0$, $\underline{v} = 30 \cos \alpha \hat{i} + 30 \sin \alpha \hat{j}$:
 $\underline{c}_1 = 30 \cos \alpha \hat{i} + 30 \sin \alpha \hat{j}$

$\therefore \underline{v} = 30 \cos \alpha \hat{i} + (30 \sin \alpha - 10t)\hat{j}$

$\underline{r} = 30t \cos \alpha \hat{i} + (30t \sin \alpha - 5t^2)\hat{j} + \underline{c}_2$

At $t=0$, $\underline{r} = 20\hat{j}$:

$\underline{c}_2 = 20\hat{j}$

$\therefore \underline{r} = 30t \cos \alpha \hat{i} + (30t \sin \alpha - 5t^2 + 20)\hat{j}$ [3]

(iii) $x = 30t \cos \alpha$ $y = 30t \sin \alpha - 5t^2 + 20$
 $t = \frac{x}{30 \cos \alpha}$ $= 30 \left(\frac{x}{30 \cos \alpha} \right) \sin \alpha - 5 \left(\frac{x}{30 \cos \alpha} \right)^2 + 20$
 $y = x \tan \alpha - \frac{x^2}{180} \sec^2 \alpha + 20$

At $x=90, y=1$: $1 = 90 \tan \alpha - 45(1 + \tan^2 \alpha) + 20$

$45 \tan^2 \alpha - 90 \tan \alpha + 26 = 0$

$\tan \alpha = \frac{90 \pm \sqrt{90^2 - 4(45)(26)}}{90}$

$= 0.3502, 1.6498$

$\alpha = 19^\circ 18', 58^\circ 47'$ [3]

Q18

A particle is projected with velocity $V \text{ ms}^{-1}$ at an angle θ above the horizontal. The horizontal and vertical displacements (in metres) from the point of projection at time t seconds are given by $x = 20\sqrt{3}t$ and $y = 20t - 5t^2$.

At time $t = 1$ the particle is moving at a speed of $S \text{ ms}^{-1}$ and is moving on a path inclined at an angle α to the horizontal such that $\alpha = \tan^{-1}(k)$.

Find the exact value of S and the time at which the particle is moving on a path inclined at an angle $\tan^{-1}\left(\frac{1}{2}k\right)$ to the horizontal.

1
3
3

$x = 20\sqrt{3}$ and $y = 20 - 10t$ If inclined at an angle $\tan^{-1}\left(\frac{1}{2}k\right)$ to the horizontal

when $t=1$ $x = 20\sqrt{3}$ and $y = 10$

$S = \sqrt{(20\sqrt{3})^2 + 10^2} = \sqrt{1300} = 10\sqrt{13} \text{ ms}^{-1}$ $\frac{20 - 10t}{20\sqrt{3}} = \frac{1}{4\sqrt{3}}$

$\tan \alpha = \frac{10}{20\sqrt{3}} = \frac{1}{2\sqrt{3}}$ $20 - 10t = 5$

$\alpha = \tan^{-1}\left(\frac{1}{2\sqrt{3}}\right)$ $\Rightarrow t = 1.5 \text{ seconds}$

$\Rightarrow k = \frac{1}{2\sqrt{3}}$

Projectile motion

Part A B C

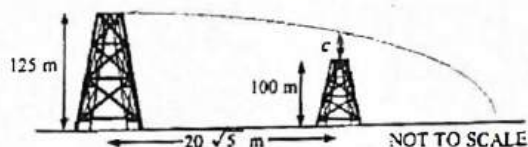
3 unit

selective

Q19

PENRITH

A projectile is shot horizontally from the top of a 125 metre tower with a velocity of V metres per second. The projectile clears a building of height 100 metres by a distance of c metres, as shown in the diagram below. The two towers are $20\sqrt{5}$ metres apart. Air resistance is ignored and use $g = 10\text{m/s}^2$.



- Derive the six equations of motion.
- Show that the displacement vector of the projectile at any time t is given by $s(t) = Vt\mathbf{i} + (125 - 5t^2)\mathbf{j}$.
- Show that $V = \frac{100}{\sqrt{25-c}}$.
- Prove that the minimum speed of projection for the projectile to just clear the 100 m tower is 20 m/s.
- Hence, find how far past the 100 metre tower the projectile will strike the ground.
- Determine the velocity and the angle of projection when it strikes the ground.

$$\dot{x} = 0$$

$$\dot{x} = \int \dot{x} dt = c,$$

$$\text{at } t=0, \theta=0^\circ, \dot{x} = V \cos \theta$$

$$\therefore \dot{x} = V \cos(0) = c,$$

$$= V$$

$$\therefore \dot{x} = V$$

$$x = \int \dot{x} dt = Vt + c_1$$

$$\text{at } t=0, x=0$$

$$x = V(0) + c_1 = 0$$

$$\therefore \dot{x} = V$$

$$\dot{y} = -10$$

$$\dot{y} = \int \dot{y} dt = -10t + c_2$$

$$\text{at } t=0, \dot{y} = V \sin \theta, \theta=0^\circ$$

$$\therefore \dot{y} = -10t$$

$$y = \int -10t dt = -5t^2 + c_4$$

$$\text{at } t=0, y=125$$

$$c_4 = 125$$

$$\therefore y = -5t^2 + 125$$

$$\text{iv) Just clear tower} \Rightarrow c=0$$

$$V = \frac{100}{\sqrt{25-0}}$$

$$= \frac{100}{5}$$

$$= 20 \text{ m/s as required.}$$

NOTE:
Done well

$$\text{v) Horizontal range} \Rightarrow y=0, V=20 \text{ m/s}$$

$$y = \frac{-5x^2}{20^2} + 125 = 0$$

$$x^2 = 10000$$

$$x = 100 \text{ m (since } x > 0)$$

$$\therefore 100 - 20\sqrt{5} \text{ m past the tower}$$

NOTE:
- Some
two
got
con-
fusi-

11) Showing some attempt at explaining displacement was horizontal and vertical component is sufficient for 1 m.

$$\text{ii) } x = Vt$$

$$\therefore t = \frac{x}{V}$$

$$y = -5t^2 + 125$$

$$= -5\left(\frac{x}{V}\right)^2 + 125$$

$$\text{at } x=20\sqrt{5} \quad y=100+c$$

$$100+c = -5\left(\frac{20\sqrt{5}}{V}\right)^2 + 125$$

$$\Rightarrow -25+c = -5\left(\frac{400 \times 5}{V^2}\right)$$

$$V^2 = \frac{10000}{25-c}$$

$$\therefore V = \frac{100}{\sqrt{25-c}} \quad (\text{since } V > 0)$$

NOTE: - Few students said $V > 0$.
- Other alternative methods were accepted.
- Not done as well.

$$\text{vi) } y=0$$

$$\Rightarrow 0 = 5t^2 - 125$$

$$125 = 5t^2$$

$$t = 5 \quad (t > 0)$$

$$\therefore \tan \theta = \frac{\dot{y}}{\dot{x}} = \frac{-10(5)}{20} = \frac{-50}{20}$$

$$\theta = \tan^{-1}\left(\frac{-50}{20}\right)$$

$$\approx 112^\circ \text{ or } 68^\circ \text{ from the ground.}$$

$$V = \sqrt{(-50)^2 + 20^2}$$

$$= 10\sqrt{29} \text{ m/s. impact velocity}$$



Projectile motion

Part A B C

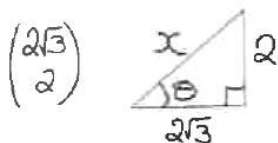
3unit

selective

Q20

A projectile has an initial velocity vector $\underline{v} = \begin{bmatrix} 2\sqrt{3} \\ 2 \end{bmatrix}$. Which of the following is the correct statement of its initial speed and angle of projection from the horizontal?

- (A) 4 m/s at 30°
(B) $\sqrt{10}$ m/s at 30°
(C) 4 m/s at 60°
(D) $\sqrt{10}$ m/s at 60°



$$\begin{aligned} 4^2 &= 2^2 + (2\sqrt{3})^2 \\ &= 4 + 12 \\ &= 16 \\ 4 &= 4 \end{aligned}$$

$$\begin{aligned} \tan \theta &= \frac{2}{2\sqrt{3}} \\ &= \frac{1}{\sqrt{3}} \\ \theta &= 30^\circ \end{aligned}$$

Q21

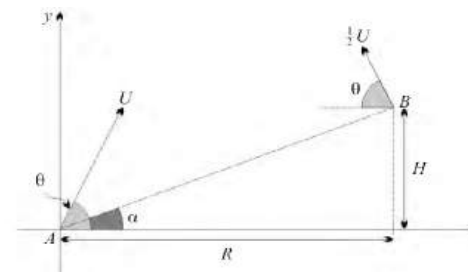
A particle moves in two dimensional space where $\underline{i}$ and $\underline{j}$ are unit vectors in the x and y directions respectively. At time t seconds its displacement from the origin is given by $\underline{r} = (6t - 4t^2)\underline{i} + 2t\underline{j}$ where all lengths are measured in metres.

- (i) Write down the particle's velocity vector in component form.
(ii) Find the speed of the particle when $t = 2$.
(iii) Show that the equation of the path of the particle is $x = 3y - y^2$.

$$\begin{aligned} \underline{r} &= 6t - 4t^2 \underline{i} + 2t \underline{j} \\ \underline{v} &= \frac{d\underline{r}}{dt} = (6 - 8t)\underline{i} + 2\underline{j} \text{ m/s} \\ \text{at } t=2, \underline{v} &= -10\underline{i} + 2\underline{j} \\ \text{speed} &= |\underline{v}| = \sqrt{(-10)^2 + 2^2} \\ &= \sqrt{104} \\ &= 2\sqrt{26} \text{ m/s} \end{aligned}$$

$$\begin{aligned} \text{(iii) } x &= 6t - 4t^2 \quad y = 2t \\ \text{from } ② \quad t &= \frac{y}{2} \\ \text{sub } t \text{ into } ① \quad x &= 6\left(\frac{y}{2}\right) - 4\left(\frac{y}{2}\right)^2 \\ x &= 3y - y^2 \end{aligned}$$

Q22



In the diagram above, points A and B are separated by a horizontal distance R and point B is located H metres higher than A. Define the angle of inclination of B from A as α . Define the origin for both motions at point A and positive directions as right and up respectively.

At the same instant, identical projectiles are launched from each location directed towards each other. The projectile from A is fired with initial speed U m/s at an angle θ above the horizontal while the object launched from B has only half as much initial speed but the same angle of elevation above the horizontal.

Consequently, the equations of motion for the projectile from A, t seconds after launch, are as follows:

$$x_A = Ut \cos \theta \quad y_A = Ut \sin \theta - \frac{1}{2}gt^2$$

Similarly for the projectile from B:

$$x_B = R - \frac{Ut \cos \theta}{2} \quad y_B = H + \frac{Ut \sin \theta}{2} - \frac{1}{2}gt^2$$

[Do NOT prove these equations.]

Given that the projectiles collide, show that this requires

$$\tan \alpha = \frac{1}{3} \tan \theta$$

$$\begin{aligned} \text{If particles collide } x_A &= x_B \text{ at time } T \\ \therefore UT \cos \theta &= R - \frac{UT \cos \theta}{2} \\ 3UT \cos \theta &= R \\ T &= \frac{2R}{3U \cos \theta} \\ \text{Also } y_A &= y_B \text{ at time } T \\ UT \sin \theta - \frac{1}{2}gT^2 &= H + \frac{UT \sin \theta}{2} - \frac{1}{2}gT^2 \\ H &= \frac{UT \sin \theta}{2} \\ &= \frac{U \sin \theta}{2} \times \frac{2R}{3U \cos \theta} \\ H &= \frac{R \tan \theta}{3} \\ \text{but } \tan \alpha &= \frac{H}{R} \\ \therefore \tan \alpha &= \frac{1}{3} \tan \theta \text{ (as required)} \end{aligned}$$

Projectile motion

Part A B C

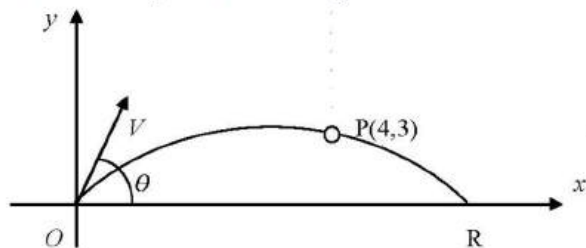
3unit

selective

Q23

TOTAL
EDUCATION

An cannon ball is projected from a point O with speed $V \text{ ms}^{-1}$ at an angle of θ to the horizontal. You are given that $V^2 = 8g$



- Assuming $\ddot{x} = 0$ and $\ddot{y} = -g$, show that $x = Vt \cos \theta$ and also find y .
 - Find the path equation of the motion
- The object passes through $P(4,3)$
- Find the angle of projection, θ .
 - Find the range of projection, R .

$$\begin{aligned} \text{(i)} \quad \ddot{x} &= 0 & \ddot{y} &= -g \\ \dot{x} &= c_1 & \dot{y} &= -gt + c_2 \\ \text{at } t=0 \Rightarrow \dot{x} &= V \cos \theta & \dot{y} &= -gt + V \sin \theta \\ x &= V \cos \theta t + D_1 & y &= -\frac{gt^2}{2} + V \sin \theta t + D_2 \\ \text{at } t=0, x=y=0 \Rightarrow D_1 &= D_2 = 0 \\ x &= V \cos \theta t \quad \text{--- (1)} & y &= -\frac{gt^2}{2} + V \sin \theta t \quad \text{--- (2)} \end{aligned}$$

$$\begin{aligned} \text{ii) (1)} \Rightarrow t &= \frac{x}{V \cos \theta} \quad \text{in (2)} \quad y = -\frac{g}{2} \left(\frac{x}{V \cos \theta} \right)^2 + V \sin \theta \cdot \frac{x}{V \cos \theta} \\ &= -\frac{g}{2} \frac{x^2}{V^2 \cos^2 \theta} + x \tan \theta \quad \text{--- (3)} \end{aligned}$$

* note: students will probably resolve the path equation as

$$\begin{aligned} y &= -\frac{g}{2} \frac{x^2}{V^2} \sec^2 \theta + x \tan \theta \\ &= -\frac{gx^2}{2V^2} (1 + \tan^2 \theta) + x \tan \theta \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \text{now } (4,3) \text{ is on path and } V^2 &= 8g \\ \Rightarrow 3 &= -\frac{g}{2} \frac{4^2}{8g \cos^2 \theta} + 4 \tan \theta \\ \Rightarrow 3 &= -\sec^2 \theta + 4 \tan \theta \\ \Rightarrow -(1 + \tan^2 \theta) + 4 \tan \theta - 3 &= 0 \\ \Rightarrow -1 - \tan^2 \theta + 4 \tan \theta - 3 &= 0 \\ \text{or } \tan^2 \theta - 4 \tan \theta + 4 &= 0 \\ \Rightarrow (\tan \theta - 2)^2 &= 0 \\ \Rightarrow \tan \theta &= 2 \\ \therefore \theta &= \tan^{-1} 2 = 63.26^\circ \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad R \text{ is between the times when } y &= 0 \\ \text{i.e. when } x=0 \text{ and } x=R \\ y=0 \Rightarrow 0 &= -\frac{gt^2}{2} + V \sin \theta t \\ \text{from (2)} \quad \Rightarrow V \sin \theta &= \frac{gt}{2} \\ t &= \frac{2V \sin \theta}{g} \\ \text{in (1)} \quad x &= \frac{V \cdot 2V \sin \theta \cos \theta}{g} = \frac{4V^2 \sin \theta \cos \theta}{g} \\ &= \frac{4 \left(\frac{2}{\sqrt{5}} \right) \left(\frac{1}{\sqrt{5}} \right)}{g} \\ &= \frac{32}{5} \\ \therefore \text{Range} &= \frac{32}{5} \text{ m.} \end{aligned}$$